Data Analysis with Python

What Data Analysis Is and the Six-Step Data Analysis Process

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Lecture Roadmap

Part I. What Is Data Analysis?

- Concept and objectives.
- Importance and applications.
- Main process overview.
- Four types of analysis.
- Tools and limitations.

Part II. Six-Step Process

- Define the problem.
- Collect data.
- Clean data.
- Analyze data.
- Visualize results.
- Interpret and decide.

Learning Outcomes

After completing this lesson, learners will be able to:

- Explain the concept and objectives of data analysis.
- Describe the role of data analysis in decision-making.
- Outline the main stages of a data analysis process.
- Distinguish descriptive, diagnostic, predictive, and prescriptive analysis.
- Recognize tools such as Excel, Python, R, Tableau, Power BI, SAS, and KNIME.
- Describe applications in business, healthcare, finance, marketing, and scientific research.
- Identify limitations related to data quality, bias, resources, tools, and context.
- Apply the concepts to simple analytical situations.

What Is Data Analysis?

Definition

Data analysis is the process of **collecting**, **cleaning**, **transforming**, and **interpreting** data to discover useful information and support decision-making.

- It converts raw data into meaningful information.
- It supports problem solving, performance evaluation, and forecasting.
- It is not only about software: it also requires a good question, appropriate methods, and contextual interpretation.

Visual Overview: What Is Data Analysis?

Data analysis overview

Data Analysis: Input, Process, Output

Component	Meaning
Input	Raw data: possibly incomplete, inconsistent, duplicated, biased, or poorly structured.
Process	Cleaning, organizing, exploring, summarizing, modeling, visualizing, and interpreting.
Output	Insights, reports, dashboards, forecasts, recommendations, or decisions.

Key message: data create value only when they are linked to a meaningful analytical question.

Quick Check: What Is Data Analysis?

Question 1

What is the main objective of data analysis?

- A. Only to store data
- B. To transform raw data into useful information
- C. To delete all unnecessary data
- D. To completely replace humans in decision-making

Question 2

Which activity is not mentioned as part of data analysis?

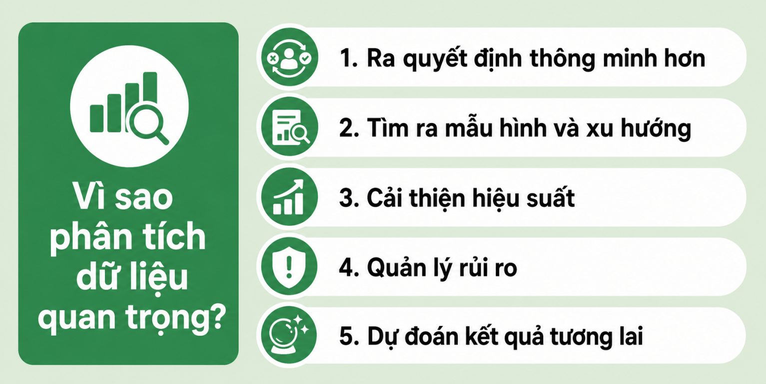
- A. Data collection
- B. Data cleaning
- C. Data interpretation
- D. Hardware manufacturing

The Importance of Data Analysis

Data analysis is important because it transforms raw information into insights that can be applied in practice.

- **Data-driven decision-making**: decisions can be based on evidence rather than intuition alone.
- **Business intelligence support**: organizations can understand customers, markets, and internal operations.
- **Performance evaluation**: processes, products, services, and strategies can be measured and improved.
- **Risk management**: potential risks can be detected, forecast, and mitigated before they become serious problems.

Illustration: Why Data Analysis Matters



Why data analysis matters

Why Data-Driven Decisions Matter

Without analysis

- Decisions rely heavily on assumptions.
- Problems may be detected too late.
- The same strategy may be applied to every group.
- Performance is difficult to evaluate objectively.

With analysis

- Decisions are supported by evidence.
- Patterns and risks can be detected earlier.
- Actions can be customized by segment.
- Performance can be tracked and compared.

Quick Check: Importance (1/2)

Question 1

Why are data-driven decisions often more reliable than decisions based only on intuition?

- A. Because data are always completely accurate
- B. Because data provide evidence to support decisions
- C. Because data eliminate all risks
- D. Because data do not require interpretation

Question 2

How does data analysis support risk management?

- A. By completely eliminating all risks
- B. By detecting and forecasting potential risks
- C. By recording risks only after they occur
- D. By replacing all control measures

Quick Check: Importance (2/2)

Question 3. True or false?

Data analysis is valuable only for large companies.

The Data Analysis Process: Overview

The data analysis process includes several major stages that transform raw data into valuable insights.

Stage	**Core role**
Define the objective	Clarify what the analysis must answer.
Collect data	Gather relevant and reliable data.
Clean and preprocess	Correct errors, missing values, duplicates, and outliers.
Exploratory analysis	Identify patterns, trends, relationships, and anomalies.
Statistical analysis	Test hypotheses, find relationships, and forecast.
Communicate results	Present insights for decision-making.

Illustration: Data Analysis Process

Overview of the data analysis process

Main Stages of the Process

1. **Define the Objective**: set clear objectives and identify the main questions.
2. **Collect Data**: collect relevant and reliable qualitative or quantitative data.
3. **Clean and Preprocess Data**: correct errors, handle missing values, remove duplicate records, and address outliers.
4. **Perform EDA**: use descriptive statistics and visualization to identify patterns, trends, relationships, and anomalies.
5. **Perform Statistical Analysis**: apply statistical methods or analytical models to test hypotheses and generate forecasts.
6. **Visualize and Communicate Results**: present findings through charts, dashboards, reports, or presentations.

Quick Check: Data Analysis Process

Question 1

What is the first stage of the data analysis process?

- A. Data visualization
- B. Model building
- C. Defining the objective
- D. Data cleaning

Question 2

Handling missing values and duplicate records belongs to which stage?

- A. Data collection
- B. Data cleaning and preprocessing
- C. Statistical analysis
- D. Communicating results

Quick Check: EDA and Process Order

Question 3

What is the main objective of exploratory data analysis?

- A. To delete the original data
- B. To identify patterns, trends, relationships, and unusual observations
- C. To create only a written report
- D. To completely replace statistical analysis

Question 4. Arrange the following activities in a logical order:

1. Clean the data
2. Define the objective
3. Visualize and communicate the results
4. Collect the data

Question 5. Case. A company wants to determine why its sales have declined. What should it do before collecting data?

Types of Data Analysis

Data analysis is commonly divided into four major types, depending on the question being addressed.

Type	**Question**	**Purpose**
Descriptive	What happened?	Summarize historical data.
Diagnostic	Why did it happen?	Identify causes and contributing factors.
Predictive	What may happen?	Estimate future outcomes or trends.
Prescriptive	What should we do?	Recommend actions or strategies.

Illustration: Four Types of Data Analysis

Four types of data analysis

Descriptive and Diagnostic Analysis

Descriptive Analysis

Descriptive analysis summarizes historical data to explain what happened. It commonly uses reports, charts, dashboards, descriptive statistics, and performance indicators.

Diagnostic Analysis

Diagnostic analysis examines data in greater depth to determine why an event or outcome occurred. It helps identify possible causes, correlations, and contributing factors behind observed patterns.

Predictive and Prescriptive Analysis

Predictive Analysis

Predictive analysis uses historical data, statistical models, and machine-learning methods to estimate what is likely to happen in the future. It supports planning by forecasting trends, risks, and opportunities.

Prescriptive Analysis

Prescriptive analysis builds on predictions to recommend actions. It supports decision-making by identifying strategies or solutions likely to produce the best outcomes.

Quick Check: Types of Data Analysis (1/2)

1. Which type of analysis answers the question "What happened?"
   - A. Descriptive analysis
   - B. Diagnostic analysis
   - C. Predictive analysis
   - D. Prescriptive analysis
2. Which type investigates causes of an observed outcome?
   - A. Descriptive
   - B. Diagnostic
   - C. Predictive
   - D. Prescriptive

Quick Check: Types of Data Analysis (2/2)

3. Forecasting sales for the next quarter belongs to which type?
   - A. Descriptive
   - B. Diagnostic
   - C. Predictive
   - D. Prescriptive
4. A system recommends changing prices to maximize profit. Which type is this?
   - A. Descriptive
   - B. Diagnostic
   - C. Predictive
   - D. Prescriptive

Matching: Core Analytical Questions

Question	**Type of analysis**
What happened?	Descriptive analysis
Why did it happen?	Diagnostic analysis
What may happen?	Predictive analysis
What should we do?	Prescriptive analysis

Data Analysis Tools

Tool	**Main use**
Microsoft Excel	Basic calculations, pivot tables, data summaries, and simple charts.
SAS	Advanced analysis, statistical modeling, and predictive analytics.
R	Statistical analysis, data modeling, and visualization.
Python	Data processing, analysis, visualization, and machine learning with Pandas, NumPy, and Matplotlib.
Tableau	Interactive dashboards and advanced data visualization.
Power BI	Business-intelligence reports and dashboards in the Microsoft ecosystem.
KNIME	Open-source workflows for data mining, processing, and machine learning.

Quick Check: Tools

1. Which tool is suitable for basic calculations, pivot tables, and simple charts?
   - A. Excel
   - B. KNIME
   - C. SAS
   - D. Tableau
2. Pandas, NumPy, and Matplotlib are commonly used with which language?
   - A. R
   - B. Python
   - C. SQL
   - D. Java
3. Which tools are commonly used to build interactive dashboards?
   - A. Tableau or Power BI
   - B. Only a word processor
   - C. A file browser
   - D. A pocket calculator

Question 4. True or false? Every data-analysis problem must use the same tool.

Applications of Data Analysis

- **Business intelligence**: monitor sales, customer behavior, operational performance, and strategy.
- **Healthcare**: monitor treatment outcomes, evaluate procedures, improve operations, and support medical research.
- **Finance**: detect fraud, evaluate risk, analyze investments, forecast markets, and support budgeting.
- **Marketing**: understand customer preferences, evaluate campaigns, segment markets, and design targeted strategies.
- **Scientific research**: interpret experimental results, test hypotheses, identify patterns, and generate knowledge.

Quick Check: Applications

1. Fraud detection is a common application of data analysis in which field?
   - A. Finance
   - B. Linguistics
   - C. Graphic design
   - D. Architecture
2. Customer segmentation and campaign-performance evaluation belong to which field?
   - A. Healthcare
   - B. Marketing
   - C. Mechanical engineering
   - D. Physics
3. In healthcare, data analysis can be used to:
   - A. Monitor treatment outcomes
   - B. Evaluate treatment effectiveness
   - C. Support medical research
   - D. All of the above

Question 4. Case. A university wants to identify factors associated with student dropout risk. In which application area does this problem belong?

Limitations of Data Analysis

- **Data-quality problems**: inaccurate, incomplete, outdated, or inconsistent data can mislead.
- **Time and resource requirements**: collection, cleaning, processing, and interpretation require effort.
- **Risk of bias**: biased datasets, methods, or assumptions may produce unreliable results.
- **Overreliance on tools**: software output can be wrong or misleading without methodological knowledge.
- **Lack of context**: quantitative data may not capture qualitative, social, behavioral, or human factors.

Quick Check: Limitations

1. What may happen when data are incomplete or inconsistent?
   - A. Results always become more accurate
   - B. Conclusions may be biased or misleading
   - C. Data cleaning becomes unnecessary
   - D. The model automatically corrects every error
2. Why should analysts not rely entirely on analytical tools?
   - A. Because every tool is inaccurate
   - B. Because results must be evaluated using methodological knowledge and context
   - C. Because tools can only be used with small datasets
   - D. Because tools cannot create charts

Question 3. True or false? A highly accurate result obtained from biased data may still lead to an inappropriate decision.

Question 4. What does the limitation "lack of context" mean?

Conclusion: What Data Analysis Provides

Summary

Data analysis provides a systematic method for transforming raw data into useful knowledge.

- Define objectives.
- Collect and prepare data.
- Explore patterns and apply analytical methods.
- Communicate results for decision-making.
- Evaluate reliability through data quality, method suitability, and correct interpretation.

End-of-Lesson Review: Part A

1. Data analysis is the process of:
   - A. Only collecting and storing data
   - B. Collecting, cleaning, transforming, and interpreting data
   - C. Only visualizing data
   - D. Only building machine-learning models
2. What should be done before collecting data?
   - A. Choose chart colors
   - B. Define the objective and analytical questions
   - C. Build the final report
   - D. Remove all outliers
3. Which type of analysis answers “Why did sales decline?”
   - A. Descriptive
   - B. Diagnostic
   - C. Predictive
   - D. Prescriptive
4. Which type answers “What might next month’s sales be?”
   - A. Descriptive
   - B. Diagnostic
   - C. Predictive
   - D. Prescriptive

End-of-Lesson Review: Part A Continued

5. Which type of analysis answers "Which strategy should the company choose?"
   - A. Descriptive
   - B. Diagnostic
   - C. Predictive
   - D. Prescriptive
6. Which of the following is a programming language widely used in data science?
   - A. Python
   - B. Tableau
   - C. Power BI
   - D. Excel
7. Missing values are usually handled during which stage?
   - A. Defining the objective
   - B. Data cleaning and preprocessing
   - C. Communicating results
   - D. Prescriptive analysis
8. Which is a limitation of data analysis?
   - A. Data always capture the full context
   - B. No expertise is needed to interpret results
   - C. Biased data can produce unreliable conclusions
   - D. Data analysis eliminates all risk

End-of-Lesson Review: True/False and Short Answers

Part B. True/False

1. Descriptive analysis focuses on explaining what happened.
2. Predictive analysis guarantees that the future will occur exactly as forecast.
3. Visualization helps communicate analytical findings to stakeholders.
4. Poor-quality data do not affect the results if powerful software is used.
5. Contextual knowledge is important when interpreting results.

Part C. Short-answer prompts

1. Present the six major stages of the data analysis process.
2. Distinguish between descriptive and diagnostic analysis.
3. Why does data quality strongly affect analytical results?
4. State two benefits and two limitations of data analysis.

Case-Based Exercises

Case 1: Sales Analysis

A store observes that June sales are 15% lower than May sales.

1. What information should descriptive analysis provide?
2. Which factors should diagnostic analysis investigate?
3. How can predictive analysis be used?
4. What recommendations might prescriptive analysis provide?

Case 2: Data Quality

A customer dataset contains many duplicate rows, missing age values, and several negative income values.

1. Which data-quality problems are present?
2. During which stage should these problems be handled?
3. What may happen if the dataset is used without cleaning?

Learning Outcomes

After completing this lesson, learners will be able to:

- Explain the role of a structured data analysis process.
- Define the problem, objectives, and success criteria of an analytical task.
- Identify and select appropriate data sources.
- Examine the structure, origin, and initial quality of a dataset.
- Handle missing values, unnecessary columns, and categorical variables.
- Use Python, Pandas, Seaborn, and Matplotlib to analyze and visualize data.
- Read and interpret correlation matrices, bar charts, histograms, and scatter plots.
- Understand basic data splitting, model training, and accuracy evaluation.
- Convert analytical results into recommendations and decisions.
- Recognize that correlation does not imply causation and that a single metric is not sufficient to judge a model.

Installing Required Libraries

```
pip install pandas seaborn matplotlib scikit-learn
```

- `pandas`: tabular data manipulation.
- `seaborn`: statistical visualization and built-in datasets.
- `matplotlib`: plotting backend and chart customization.
- `scikit-learn`: train-validation split, models, and evaluation metrics.

Why Use a Structured Process?

Data analysis follows a structured approach in which:

- **Step-by-step process**: raw data are transformed into meaningful insights.
- **Systematic approach**: the process helps improve accuracy and reliability.
- **Better decision-making**: decisions are based on evidence rather than intuition alone.

Quick Check

1. What is the main objective of a structured data analysis process?
2. True or false? A systematic data analysis process helps improve the accuracy and reliability of results.

The Six-Step Workflow

Step	Central question
Define the problem	What problem must be solved?
Collect data	What data are needed and where can they be obtained?
Clean the data	Are the data ready for analysis?
Analyze the data	What do the data show?
Visualize the results	How can the results be made easier to understand?
Interpret and decide	What do the results mean and what should be done next?

Illustration: Six-Step Workflow

Structured data analysis workflow

Step 1: Define the Problem

Before beginning any analytical activity, the problem must be clearly defined.

- Clarify the question, objective, or analytical opportunity.
- Ensure the analytical goal aligns with stakeholder expectations.
- Keep the analysis focused and relevant.
- Avoid unnecessary data collection.

Main tasks

Identify the core problem, define expected outcomes, understand context and constraints, and define success criteria.

Step 1 Example: Sales Decline

A company notices that sales have declined significantly over the last three months.

Instead of asking only "Why did sales decline?", ask:

- Which products experienced the largest decline?
- Which region had the greatest decrease?
- Which customer groups were most affected?
- Is the decline related to pricing, inventory, or marketing activities?

Quick Check: Step 1

1. Why should the problem be defined before data collection?
   - A. To reduce file size
   - B. To keep the analysis focused and relevant
   - C. To avoid using charts
   - D. To remove all qualitative data
2. Which activity belongs to the problem-definition step?
   - A. Handling missing values
   - B. Defining success criteria
   - C. Training a model
   - D. Drawing a heatmap

Question 3. Case. A university wants to investigate why students drop out. Suggest two more specific analytical questions.

Step 2: Collect Data

After defining the problem, collect data from appropriate sources.

- Data may come from internal databases, APIs, surveys, web scraping, or public datasets.
- Data must be relevant, accurate, and sufficiently complete.
- Combining multiple sources may enrich the analysis.
- The origin and structure of each dataset should be documented for transparency.
- Update frequency, file format, and refresh requirements should be considered.

Step 2 Example: Titanic Dataset

```
import seaborn as sns
import pandas as pd

titanic = sns.load_dataset("titanic")
titanic.head()
```

- The Titanic dataset is built into Seaborn.
- `head()` displays the first few rows.
- The dataset is used to demonstrate cleaning, analysis, visualization, and model building.

Illustration: Titanic Dataset Preview

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Titanic dataset preview

Quick Check: Step 2

1. Which of the following can be used as a data source?
   - A. Internal databases
   - B. APIs
   - C. Surveys
   - D. All of the above
2. Why should the origin of a dataset be documented?
   - A. To improve transparency and auditability
   - B. To make the dataset larger
   - C. To avoid data cleaning
   - D. To replace data analysis
3. Which function is used to load the Titanic dataset in the example?
   - A. `pd.read_csv()`
   - B. `sns.load_dataset()`
   - C. `plt.load()`
   - D. `np.loadtxt()`

Step 3: Clean the Data

Raw data are rarely ready for direct analysis.

Data cleaning includes

- Handling missing values.
- Removing duplicate records.
- Standardizing formats.
- Converting categorical variables into numerical form.
- Removing irrelevant, redundant, or inconsistent columns.

Well-prepared data improve the reliability and accuracy of analytical results.

Step 3: Check and Handle Missing Data

```
print(titanic.isnull().sum())

titanic["age"].fillna(
    titanic["age"].median(),
    inplace=True
)

titanic["embarked"].fillna(
    titanic["embarked"].mode()[0],
    inplace=True
)
```

- `isnull().sum()` counts missing values in each column.
- Missing values in `age` are replaced by the median.
- Missing values in `embarked` are replaced by the most frequent value.

Step 3: Remove Columns and Encode Categories

```
titanic.drop(
    ["deck", "embark_town", "alive", "class", "who", "adult_male"],
    axis=1,
    inplace=True
)

titanic["sex"] = titanic["sex"].map({"male": 0, "female": 1})
titanic["embarked"] = titanic["embarked"].map({"C": 0, "Q": 1, "S": 2})

titanic.head()
```

Categorical variables often need to be converted before analysis or modeling.

Illustration: Cleaned Titanic Data

	survived	pclass	sex	age	sibsp	parch	fare	embarked	alone
0	0	3	0	22.0	1	0	7.2500	2	False
1	1	1	1	38.0	1	0	71.2833	0	False
2	1	3	1	26.0	0	0	7.9250	2	True
3	1	1	1	35.0	1	0	53.1000	2	False
4	0	3	0	35.0	0	0	8.0500	2	True

Titanic data after cleaning and encoding

Quick Check: Step 3

1. Which command counts missing values in each column?
   - A. `titanic.head()`
   - B. `titanic.isnull().sum()`
   - C. `titanic.describe()`
   - D. `titanic.drop()`
2. In the example, missing values in age are replaced by:
   - A. The minimum value
   - B. The maximum value
   - C. The median
   - D. Zero
3. Why is the `sex` column converted into numerical values?
   - A. To reduce rows
   - B. To make the data more suitable for analytical algorithms or models
   - C. To change gender
   - D. To create a pie chart

Question 4. True or false? Every column containing missing values can be removed without considering context.

Step 4: Analyze the Data

Data analysis is the core step in which the analyst searches for patterns, trends, and relationships.

- Calculate measures such as mean, median, mode, and variance.
- Identify correlations, trends, and unusual observations.
- Apply models such as regression, clustering, or classification.
- Compare results with expectations or hypotheses.

Step 4: Correlation Matrix

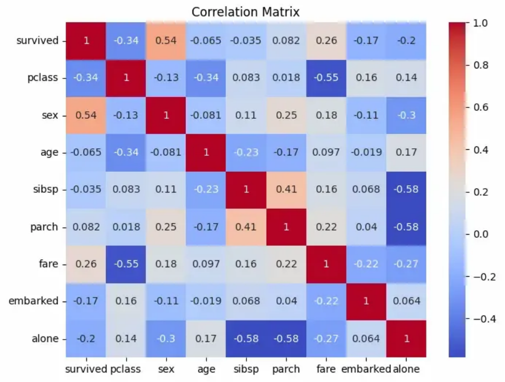
```python
import matplotlib.pyplot as plt

plt.figure(figsize=(8, 6))
sns.heatmap(
    titanic.corr(),
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Matrix")
plt.show()
```

A heatmap helps observe the degree of correlation among numerical variables.

Illustration: Correlation Matrix



Correlation matrix for the Titanic dataset

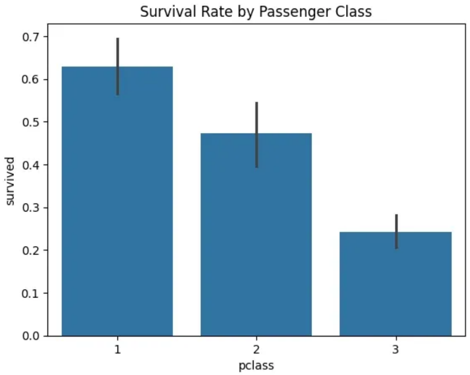
Step 4: Survival Rate by Passenger Class

```
sns.barplot(
    x="pclass",
    y="survived",
    data=titanic
)

plt.title("Survival Rate by Passenger Class")
plt.show()
```

The bar chart compares the average survival rate across passenger classes.

Illustration: Survival Rate by Passenger Class



Survival rate by passenger class

Quick Check: Step 4

1. What is a correlation matrix used for?
   - A. Checking file types
   - B. Examining relationships among variables
   - C. Removing duplicates
   - D. Loading data from an API
2. In the example, the barplot compares:
   - A. Age and fare
   - B. Passenger class and survival rate
   - C. Gender and port
   - D. Number of columns and rows
3. Which models may be used during data analysis?
   - A. Regression
   - B. Clustering
   - C. Classification
   - D. All of the above

Question 4. Case. If two variables are highly correlated, can we conclude with certainty that one causes the other?

Step 5: Visualize the Results

Visualization makes complex data easier to understand.

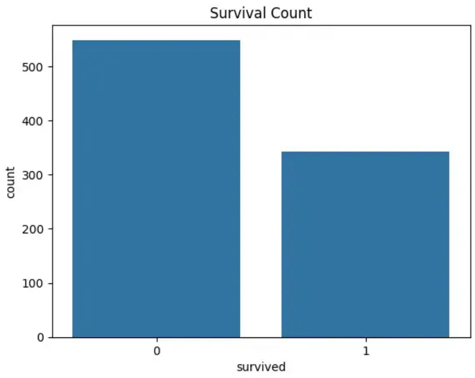
- Charts, graphs, and dashboards highlight important trends, patterns, and unusual observations.
- A good visualization should be clear, intuitive, and useful for decision-making.
- Select chart types according to analytical questions: histogram, scatter plot, bar chart, heatmap, etc.
- Avoid excessive detail that distracts from the main message.

Step 5: Count Plot and Age Distribution

```
sns.countplot(
    x="survived",
    data=titanic
)
plt.title("Number of Passengers by Survival Status")
plt.show()

sns.histplot(
    titanic["age"],
    kde=True
)
plt.title("Age Distribution")
plt.show()
```

Illustration: Survival Count and Age Distribution



Passenger counts by survival status

Age distribution of Titanic passengers

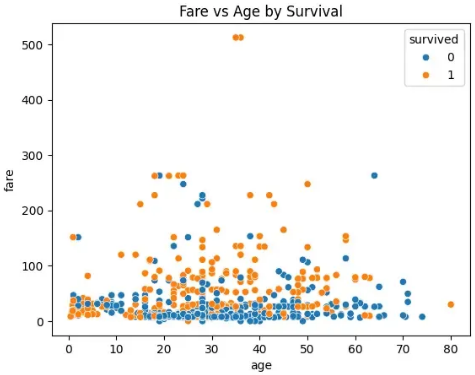
Step 5: Scatter Plot of Age and Fare

```
sns.scatterplot(
    x="age",
    y="fare",
    hue="survived",
    data=titanic
)

plt.title("Relationship Between Fare and Age by Survival Status")
plt.show()
```

The `hue` argument colors points by survival status.

Illustration: Fare vs Age by Survival Status



Age and fare by survival status

Quick Check: Step 5

1. Which chart is suitable for displaying the distribution of age?
   - A. Histogram
   - B. Network chart
   - C. Gantt chart
   - D. Geographic map
2. What is the purpose of `hue="survived"` in the scatter plot?
   - A. To remove the `survived` column
   - B. To distinguish points by survival status
   - C. To change the figure size
   - D. To calculate the mean

Question 3. True or false? A chart is always more effective when it contains more details.

Question 4. Why should the chart type be selected according to the analytical question?

Step 6: Interpret Results and Make Decisions

The final step is to transform analytical results into actionable insights.

- Place results in the context of the original problem.
- Provide actionable recommendations based on findings.
- Communicate results clearly to stakeholders.
- Monitor outcomes and repeat the process for continuous improvement.

Interpretation connects the numerical result with practical decision-making.

Step 6: Building a Survival Prediction Model

```python
X = titanic.drop("survived", axis=1)
y = titanic["survived"]

X_train, X_val, y_train, y_val = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_val)
accuracy = accuracy_score(y_val, y_pred)
print(f"Model accuracy: {accuracy:.4f}")
```

Step 6: Interpreting Accuracy

Result

```
Model accuracy: 0.8101
```

- The model correctly predicts approximately 81.01% of observations in the validation set.
- Accuracy alone should not be used to evaluate a model.
- Practical applications may require additional metrics and careful context-specific interpretation.
- The code uses `train_test_split`, `RandomForestClassifier`, and `accuracy_score`, which must be imported from Scikit-learn.

Quick Check: Step 6

1. What is the objective of the interpretation step?
   - A. Only to present numbers
   - B. To transform analytical results into insights and actions
   - C. To delete all data
   - D. To change the original problem
2. An accuracy value of `0.8101` is approximately:
   - A. 8.101%
   - B. 18.01%
   - C. 81.01%
   - D. 810.1%
3. Why should results be monitored after a decision is implemented?
   - A. To determine whether the decision is effective and make adjustments when needed
   - B. To increase the number of columns
   - C. To avoid communicating with stakeholders
   - D. To ensure no further analysis is needed

Question 4. True or false? Accuracy alone is always sufficient to fully evaluate a classification model.

Process Summary

Step	Main content	Central question
Define the problem	Objective, context, success criteria	What problem must be solved?
Collect data	Relevant sources	What data are needed?
Clean data	Missing values, formatting	Are data ready?
Analyze data	Patterns and relationships	What do data show?
Visualize results	Charts and dashboards	How to make results clear?
Interpret and decide	Recommendations and actions	What should be done?

Review: Multiple-Choice Questions 1-4

1. What is the first step in the data analysis process?
   - A. Clean the data
   - B. Define the problem
   - C. Visualize the data
   - D. Train a model
2. Which activity belongs to the data-collection step?
   - A. Identify data sources
   - B. Handle missing values
   - C. Calculate a correlation matrix
   - D. Evaluate model accuracy
3. The median is commonly used to:
   - A. Rename a column
   - B. Fill missing values in a numerical variable
   - C. Create an API
   - D. Draw a scatter plot
4. Which function is used to visualize a correlation matrix in the example?
   - A. `sns.heatmap()`
   - B. `sns.load_dataset()`
   - C. `pd.DataFrame()`
   - D. `model.fit()`

Review: Multiple-Choice Questions 5-8

5. Which chart is used to show the age distribution?
   - A. Histogram
   - B. Pie chart
   - C. Gantt chart
   - D. Geographic heatmap
6. What does `test_size=0.2` mean?
   - A. 20% of data are used as validation
   - B. 20% are deleted
   - C. There are 20 input variables
   - D. Model has 20% accuracy
7. An accuracy of `0.8101` is equal to:
   - A. 0.8101%
   - B. 8.101%
   - C. 81.01%
   - D. 810.1%
8. Which step converts analytical results into recommendations?
   - A. Data collection
   - B. Data cleaning
   - C. Interpretation and decision-making
   - D. Visualization only

Review: True/False

Part B. True/False

1. The analytical objective should be defined before data collection.
2. Raw data can always be used directly to build a model.
3. A correlation matrix can help identify relationships among variables.
4. A high correlation always proves causation.
5. Data visualization can support communication with stakeholders.
6. After a decision has been made, there is no need to monitor the outcome.

Review: Short Answers

Part C. Short answers

1. Present the six main steps.
2. Why is defining the problem important?
3. State three common problems in raw data.
4. Distinguish data analysis and visualization.
5. Why must model results be interpreted in context?

Practical Exercises

Exercise 1. Explore the Titanic Dataset

Display the first five rows, determine rows and columns, check data types, count missing values, and comment on quality.

Exercise 2. Clean the Data

Fill missing values, remove unnecessary columns, convert `sex` into numerical values, and recheck the data.

Exercise 3. Analyze and Visualize

Calculate descriptive statistics, plot age distribution, compare survival by class, draw age/fare scatter plot, and write observations.

Practical Exercise 4: Interpret the Result

Assume that a survival-prediction model achieves an accuracy of 81.01%.

1. Explain what the result means.
2. State two reasons why accuracy alone should not be used.
3. Suggest two additional evaluation metrics.
4. State one ethical or practical limitation of using the model.

Answer Key: What Is Data Analysis?

- Quick Check - What Is Data Analysis: Q1 B; Q2 D.
- Importance: Q1 B; Q2 B; Q3 False.
- Data Analysis Process: Q1 C; Q2 B; Q3 B; Q4 order is $2 \rightarrow 4 \rightarrow 1 \rightarrow 3$.
- Types: Q1 A; Q2 B; Q3 C; Q4 D.
- Tools: Q1 A; Q2 B; Q3 A; Q4 False.
- Applications: Q1 A; Q2 B; Q3 D; Q4 Education.
- Limitations: Q1 B; Q2 B; Q3 True; Q4 quantitative data may miss qualitative, social, behavioral, or human context.

Answer Key: Review for Part I

- Multiple choice: 1 B; 2 B; 3 B; 4 C; 5 D; 6 A; 7 B; 8 C.
- True/False: 1 True; 2 False; 3 True; 4 False; 5 True.
- Short answer Q1: define objective, collect data, clean/preprocess data, perform EDA, perform statistical analysis, visualize/communicate results.
- Short answer Q2: descriptive explains what happened; diagnostic investigates why it happened.
- Short answer Q3: poor-quality data can produce misleading patterns, biased conclusions, and unreliable models.
- Short answer Q4: benefits include better decisions and risk management; limitations include bias, poor quality, resource needs, and lack of context.

Answer Key: Six-Step Process Quick Checks

- Introduction: Q1 B; Q2 True.
- Step 1: Q1 B; Q2 B; case examples: student groups with highest dropout rates; relationship with grades, finances, or engagement.
- Step 2: Q1 D; Q2 A; Q3 B.
- Step 3: Q1 B; Q2 C; Q3 B; Q4 False.
- Step 4: Q1 B; Q2 B; Q3 D; Q4 No - correlation does not imply causation.
- Step 5: Q1 A; Q2 B; Q3 False; Q4 different chart types fit different data and communication objectives.
- Step 6: Q1 B; Q2 C; Q3 A; Q4 False.

Answer Key: Review for Part II

- Multiple choice: 1 B; 2 A; 3 B; 4 A; 5 A; 6 A; 7 C; 8 C.
- True/False: 1 True; 2 False; 3 True; 4 False; 5 True; 6 False.
- Short answer Q1: define problem, collect data, clean data, analyze data, visualize results, interpret and decide.
- Q2: defining the problem determines the objective, scope, required data, and success criteria.
- Q3: common raw-data problems include missing values, duplicates, inconsistent formats, outliers, and unsuitable data types.
- Q4: analysis discovers patterns and draws conclusions; visualization presents data or findings visually.
- Q5: context determines what a statistical or model result actually means for decisions.

Closing Message

Core idea

Good data analysis is not a single tool or chart. It is a disciplined process that starts with a clear question, prepares data carefully, explores patterns, communicates findings, and connects results to decisions.

- Use the process to avoid jumping directly into tools.
- Use quizzes to check conceptual understanding.
- Use exercises to build practical fluency with real datasets.